

Take-Home Assignment for Segment 06

Using a FASTA file with protein sequences (amino acids this time instead of nucleotides) with all proteins of an organism, write a program that lists the top 15 proteins in terms of percentage of hydrophobic amino acids. Print the protein description (the header from the FASTA file), length, and percentage of hydrophobic amino acids. Exclude small peptides (<100 amino acids). Using your knowledge of biology (and biochemistry), what types of proteins would you expect to contain many hydrophobic amino acids? Are your results consistent with the expectation?

For the assignment, run your program for at least two prokaryotic genomes and provide your code and a brief conclusion as to whether the proteins with most hydrophobic amino acids are what you would expect.

Do not use any system-provided 'sort' function in your code.

Maximum award for completing the assignment is 15 points (12 for the program, 3 for the presentation and evaluation of the results).

Notes and useful resources:

- For the purpose of this assignment, consider L, I, V, F, M to be hydrophobic. If you are not familiar with amino acid code, see <https://www.bioinformatics.org/sms/iupac.html>.
- You can download the required files from GenBank. Go to <https://www.ncbi.nlm.nih.gov/> -- it is the same as downloading gene sequences for the previous assignment, only choose "Protein (FASTA)" instead of coding sequences.
- Recommended approach for picking the top 15 proteins:
 1. Define a structure to store the hydrophobic amino acid content, name (header), and length of a protein.
 2. Create an array with 16 elements of the structure (from [0] to [15]). You can make things easier by setting the hydrophobic amino acid content in all elements of the array to some negative number (lower than any actual protein can have).
 3. Write the main loop to read the file and calculate the hydrophobic amino acid content for each protein; fill the structure with index 15 with the protein name, length, and hydrophobic amino acid content.
 4. Write another loop inside the main loop going backwards from 15 to 1 and if the element i has higher hydrophobic amino acid content than the element i-1, swap them (keep in mind that you need to use a temporary placeholder to swap two variables; for example, to swap A and B, you need to do Temp=A; A=B; B=Temp).
- If you need to find information about a protein, UniProt can be a useful resource (<https://www.uniprot.org/>).
- If you struggle, ask for help.

How to use an array of type struct:

```
// define the structure:
struct ProtStruct {
    double HPhob;
    string Name;
    long Length;
};
...
// Create the array:
ProtStruct Array[16];
...
// to swap elements i and i-1 if the HPhob in the element i is higher than in element i-1:
If (Array[i]. HPhob >Array[i-1]. HPhob) {
    ProtStruct Temp=Array[i];
    Array[i]=Array[i-1];
    Array[i-1]=Temp;
}
...
// in the end, elements 0-14 should contain the top 15 proteins by hydrophobic amino acid content
// try to find a more fitting name for the array than "Array"
```